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Introducing OpenShift Service Mesh

Lesson

Lab Environment

Guided Exercise: Service Mesh Showroom Application

Describe how the Bookinfo application provides a realistic scenario for demonstrating OpenShift Service Mesh features, including traffic routing, observability, and security.

Outcomes

- Explore OpenShift Service Mesh (OSSM) features by using the Bookinfo application.
- Access the Bookinfo application through the OSSM ingress gateway.
- Navigate with the OpenShift web console to view the service mesh topology.
- Generate traffic to observe load balancing across service versions.
- Use Service Mesh traffic graphs to visualize microservice communication.
- Configure traffic shifting to route 100% traffic to a specific service version.
- Verify traffic routing changes in the application interface.

As the student user on the workstation machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available.

```
[student@workstation ~]$ lab start meshintro-bookinfo
```

Also, run the following command to update your `$PATH` variable and make the `traffic_gen.py` command available immediately. Run it once after creating a new environment.

```
[student@workstation ~]$ source ~/.bashrc
```

The command `lab start` ensures that OpenShift Service Mesh is deployed and configured on the OpenShift cluster in your lab environment, and deploys all the components of the Bookinfo application.

Instructions

1. Verify the initial status of the service mesh.

- 1.1. In the workstation machine, open the OpenShift web console in your browser.

Go to <https://console-openshift-console.apps.ocp4.example.com>, click `htpasswd_provider` and use `developer` as the user and the password. If prompted about the guided tour, then click `Skip tour`.

- 1.2. From the side menu, click the **Developer** menu item to ensure that you use the developer perspective.



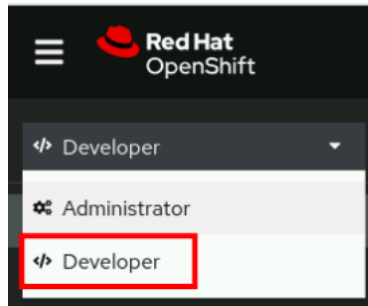


Figure 1.10: OpenShift Developer perspective navigation menu

- 1.3. Click the **Topology** menu item to open the Topology view and inspect the deployments. Ensure that the `meshintro-bookinfo` project is selected in the **Projects** list dropdown.

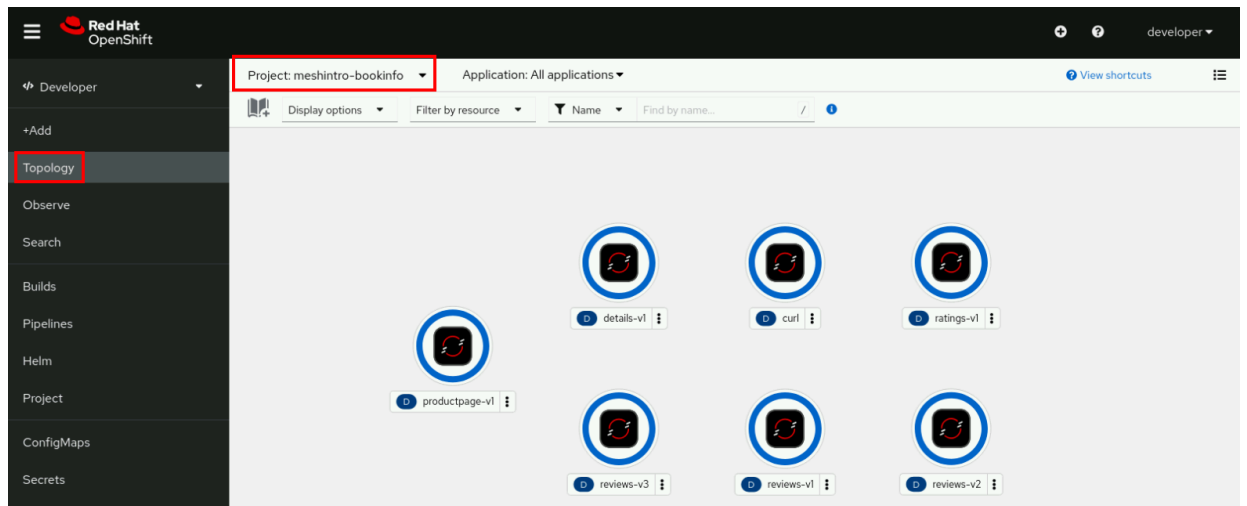


Figure 1.11: Bookinfo application topology view

Note that there are seven deployments in the project, and none have a route icon in the upper-right corner. This is because all applications are only accessible externally through the OSSM ingress gateway.

2. Explore the Bookinfo application.

- 2.1. Open the Bookinfo `productpage` page in a new tab in your browser.

Go to `http://istio-ingressgateway-istio-ingress.apps.ocp4.example.com/productpage`.

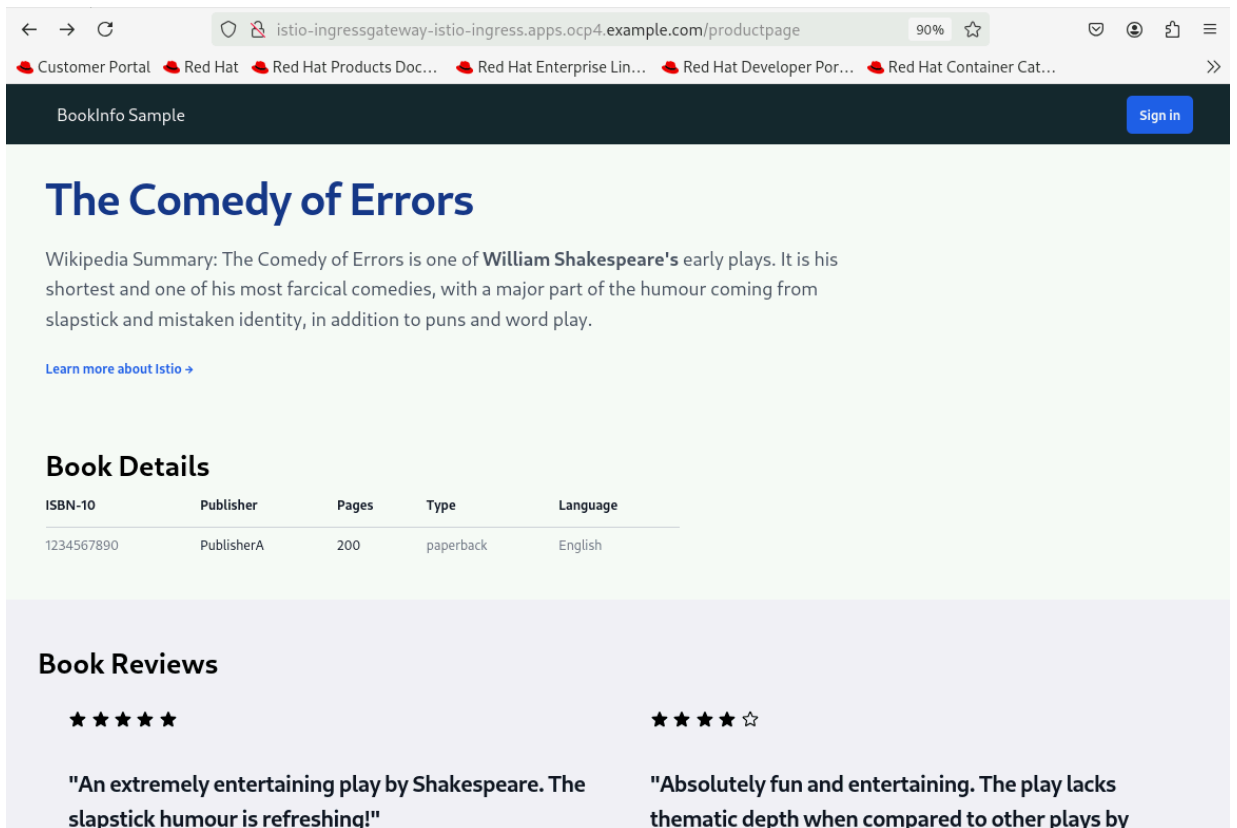


Figure 1.12: Bookinfo application homepage

Observe the star ratings under the Book Reviews section. Refresh the page multiple times and observe that the stars change between red, black, and no stars in approximately equal distribution. This variation occurs because the reviews service has three different versions:

- `reviews v1`: no stars
- `reviews v2`: black stars
- `reviews v3`: red stars

- 2.2. In a new terminal window, log in as the developer user and the developer password, switch to the meshintro-bookinfo project:

```
[student@workstation meshintro-bookinfo]$ oc login -u developer \
-p developer https://api.ocp4.example.com:6443
Login successful.
...output omitted...
[student@workstation meshintro-bookinfo]$ oc project meshintro-bookinfo
...output omitted...
```

- 2.3. Go to the exercise directory and execute the `traffic_gen.py` script to generate traffic for the reviews Bookinfo service. Press `Ctrl+C` after some requests to interrupt the script.

```
[student@workstation ~]$ cd ~/course/labs/meshintro-bookinfo
[student@workstation meshintro-bookinfo]$ traffic_gen.py continuous.yaml
Continuous mode: 600s to http://istio-ingressgateway-istio-ingress.apps.ocp4.example.com/reviews/1
curl -s http://istio-ingressgateway-istio-ingress.apps.ocp4.example.com/reviews/1
[1] [✓] HTTP 200 -- red (0.0s)
[2] [✓] HTTP 200 -- No stars (0.1s)
[3] [✓] HTTP 200 -- black (0.4s)
[4] [✓] HTTP 200 -- black (0.5s)
[5] [✓] HTTP 200 -- black (0.6s)
[6] [✓] HTTP 200 -- No stars (0.8s)
[7] [✓] HTTP 200 -- No stars (0.9s)
[8] [✓] HTTP 200 -- red (1.0s)
[9] [✓] HTTP 200 -- black (1.1s)
...output omitted...
Interrupted by user (Ctrl+C). Stopping continuous mode...
```

Traffic Statistics

Total Request	Success Rate	Average	P50	P95
39	100.0% (39/39)	23.1ms	16.0ms	79.8ms

Response Distribution

Response	Count	Percentage
No stars	15	38.5%
black	14	35.9%
red	10	25.6%

Observe that all three versions of the reviews services respond in a balanced way. Your values might be different.

NOTE

Remember to run `source ~/.bashrc` to update your `$PATH` variable and make the `traffic_gen.py` script available.

```
[student@workstation ~]$ source ~/.bashrc
```

- 2.4. Re-execute the `traffic_gen.py` script to generate traffic for the reviews Bookinfo service. Leave the script running.

```
[student@workstation meshintro-bookinfo]$ traffic_gen.py continuous.yaml
...output omitted...
```

3. Explore the traffic flows between the microservices.
 - 3.1. Go to the OpenShift web console in your browser and switch to the **Administrator** perspective. In the upper-left corner, click **Developer** and select **Administrator**.
 - 3.2. Go to the OSSM overview section by clicking **Service Mesh > Overview** menu entry.

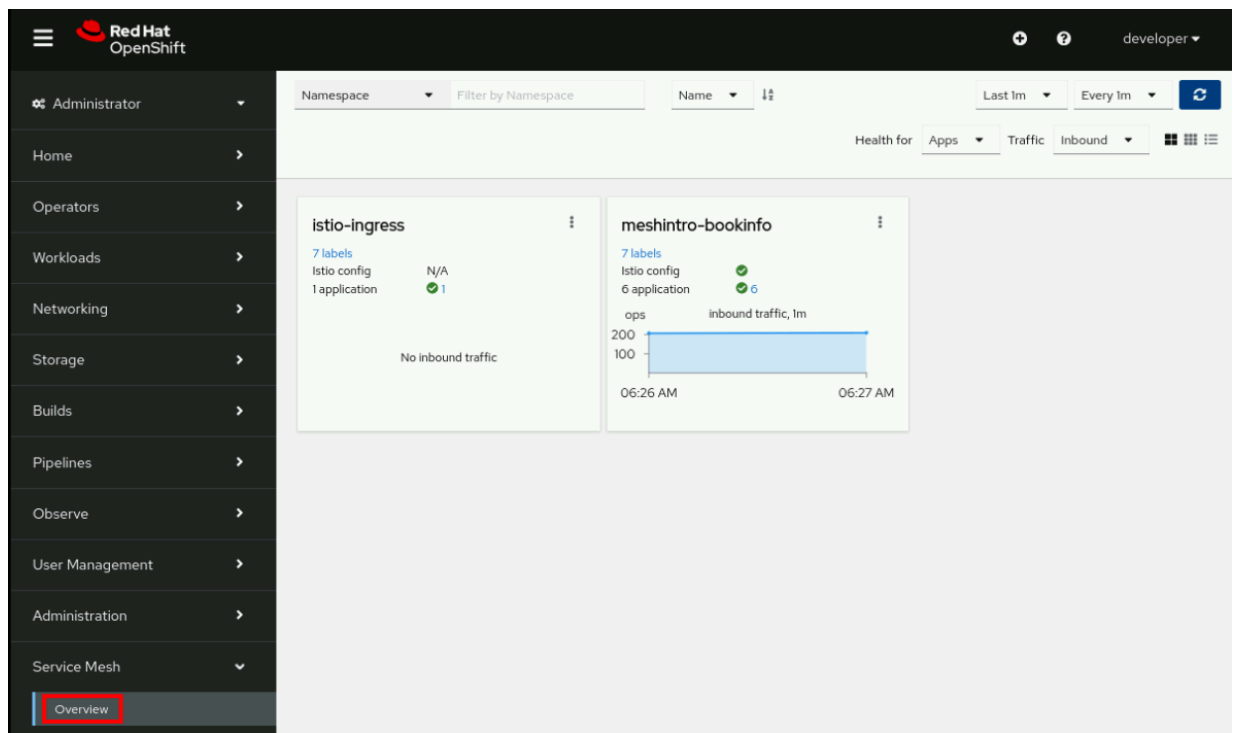


Figure 1.13: Service Mesh overview perspective

Note that the meshintro-bookinfo project has an average of about two hundred requests per minute.

NOTE

Your requests per minute statistic might appear differently.

- 3.3. Click Traffic Graph menu entry and in the Select Namespaces combobox at the top, select only the meshintro-bookinfo project.

NOTE

You must click outside the selection menu before the filter is applied.

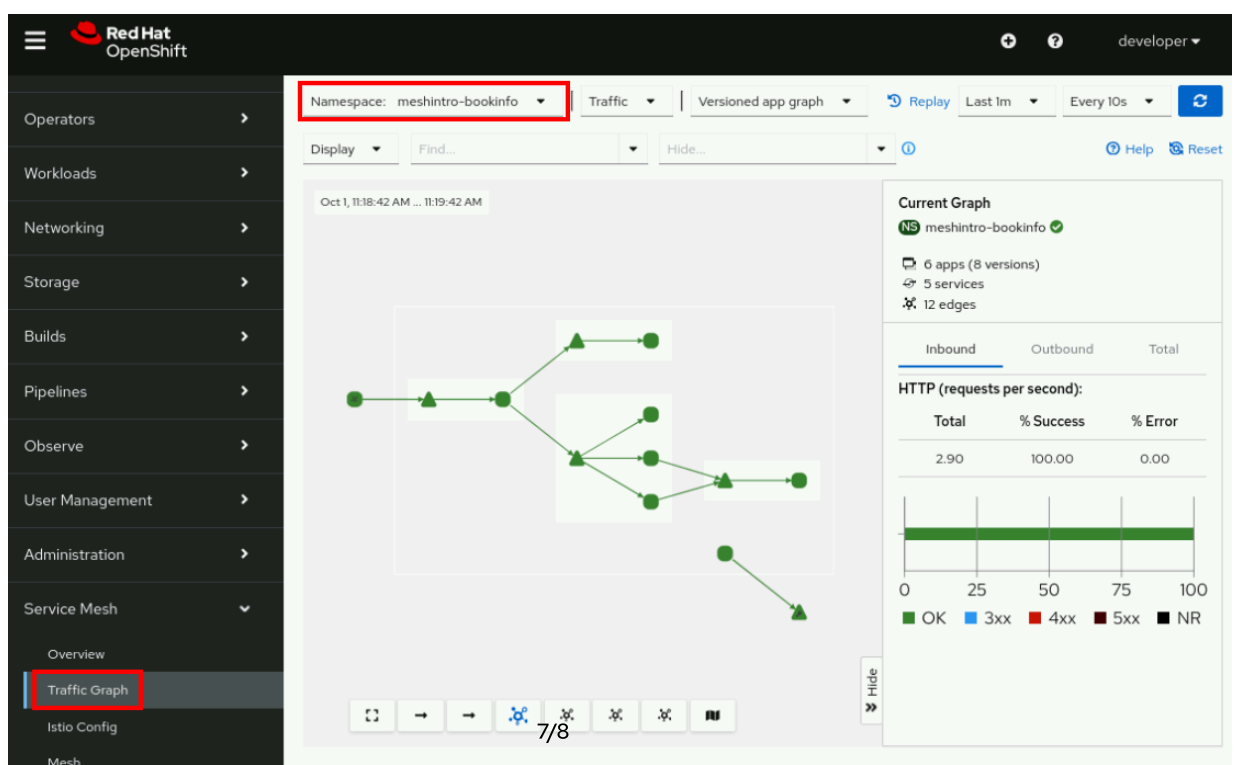


Figure 1.14: Service Mesh traffic graph perspective

Observe the traffic flow graph for the project.

NOTE

Note that your graph may look different. Different settings and scenarios can alter the exhibition of this and other graphs in the course. Settings such as zoom level, traffic metrics per refresh, and more.

- 3.4. Hover over the triangles with your mouse to view the Bookinfo services. For example, hover your mouse over the first triangle from the left, which is the `productpage` service.

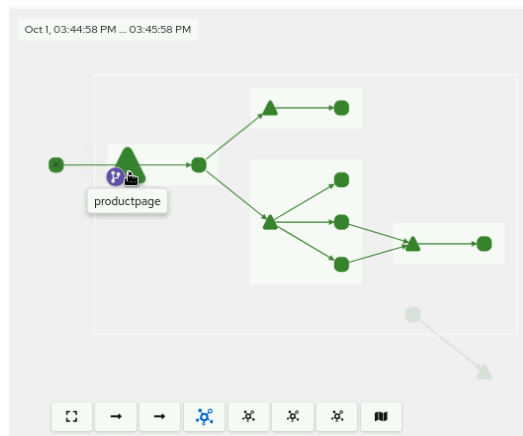


Figure 1.15:
productpage service details in traffic graph

- 3.5. Following its outbound arrows, hover the mouse over the topmost triangle. This is the `details` service that provides the book information.

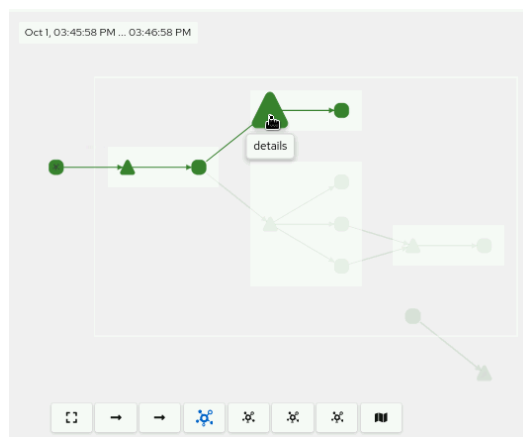


Figure 1.16:
details service information in traffic graph

- 3.6. Below it, you can see the `reviews` service. Note that there are three outgoing arrows from the `reviews` triangle. Those are the three different versions of the `reviews` application.

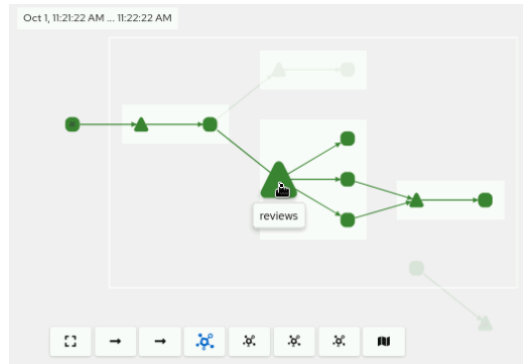


Figure 1.17:
reviews service with three version variants in traffic graph

- 3.7. By hovering over the rounded squares, you can see the v1, v2, and v3 versions of the reviews service. Note that v1 does not invoke the ratings service on its right. Only v2 and v3 invoke the ratings service.
4. Change the traffic flow to a fixed version of the reviews microservice.
 - 4.1. Click the reviews service in the graph to select it. In the right panel that opens, expand the ellipsis (:) menu and click the **Traffic Shifting** menu item.

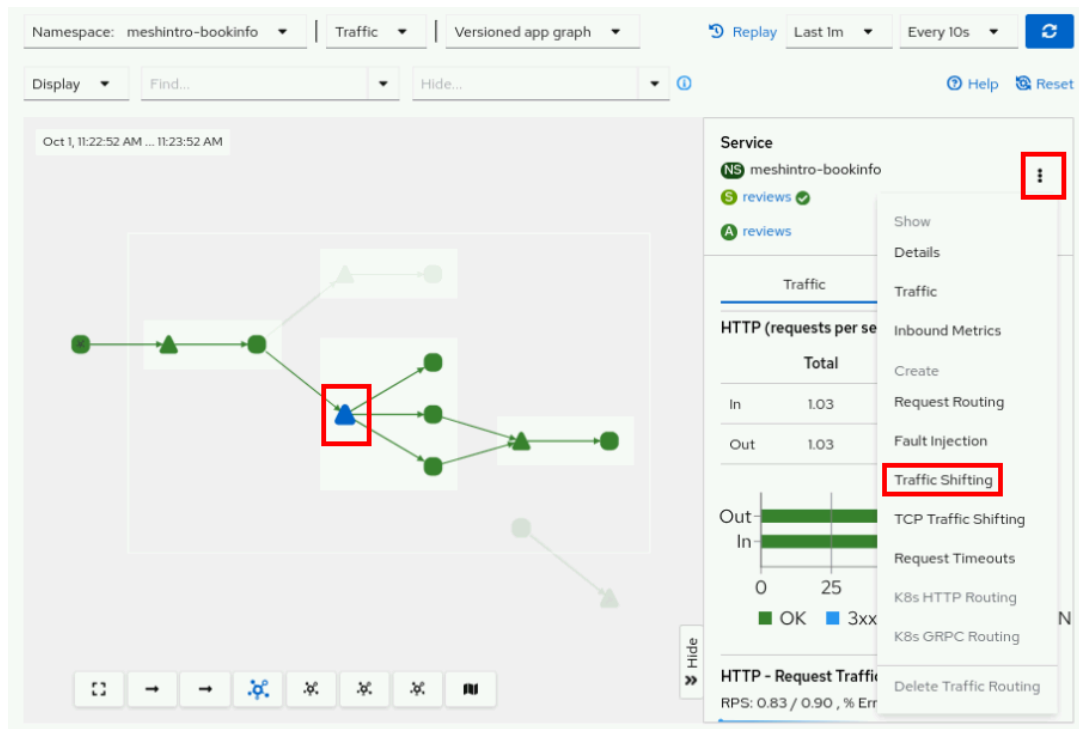


Figure 1.18:
Add traffic shifting configurations to the reviews service

- 4.2. In the Create Traffic Shifting form, define the reviews-v3 **Traffic Weight** to 100% and reviews-v1 and reviews-v2 to 0%. Then click **Preview** to visualize the OSSM resources to be created.

Figure 1.19:
reviews service traffic weight configuration form

- 4.3. Select the VirtualService tab, scroll down, and ensure that the weight settings are such that 100% of traffic goes to service subset v3 and 0% to v1 and v2. Then click Create.

Figure 1.20:
reviews VirtualService traffic routing configuration

- 4.4. On the confirmation dialog, click Create to confirm the OSSM resources creation.
- 4.5. Open the terminal window where the traffic_gen.py script is running. Notice that only the reviews v3 service version, the red stars, is responding.

```
...output omitted...
[4245] ✓ HTTP 200 -- red (483.1s)
[4246] ✓ HTTP 200 -- red (483.2s)
[4247] ✓ HTTP 200 -- black (483.3s)
[4248] ✓ HTTP 200 -- No stars (483.4s)
[4249] ✓ HTTP 200 -- No stars (483.5s)
[4250] ✓ HTTP 200 -- No stars (483.6s)
[4251] ✓ HTTP 200 -- red (483.7s) ❶
[4252] ✓ HTTP 200 -- red (483.8s)
[4253] ✓ HTTP 200 -- red (484.0s)
[4254] ✓ HTTP 200 -- red (484.1s)
[4255] ✓ HTTP 200 -- red (484.2s)
[4256] ✓ HTTP 200 -- red (484.3s)
[4257] ✓ HTTP 200 -- red (484.4s)
[4258] ✓ HTTP 200 -- red (484.5s)
[4259] ✓ HTTP 200 -- red (484.6s)
[4260] ✓ HTTP 200 -- red (484.7s)
[4261] ✓ HTTP 200 -- red (484.8s)
[4262] ✓ HTTP 200 -- red (485.0s)
[4263] ✓ HTTP 200 -- red (485.1s)
...output omitted...
```

❶ The instant OpenShift Service Mesh applied the traffic shifting configuration.

NOTE

If your `traffic_gen.py` finished, then you can relaunch it with the following command:

```
[student@workstation ~]$ cd ~/course/labs/meshintro-bookinfo
[student@workstation meshintro-bookinfo]$ traffic_gen.py continuous.yaml
```

- 4.6. In a new web browser tab, open the `productpage` application by going to the istio ingress gateway URL `http://istio-ingressgateway-istio-ingress.apps.ocp4.example.com/productpage`.

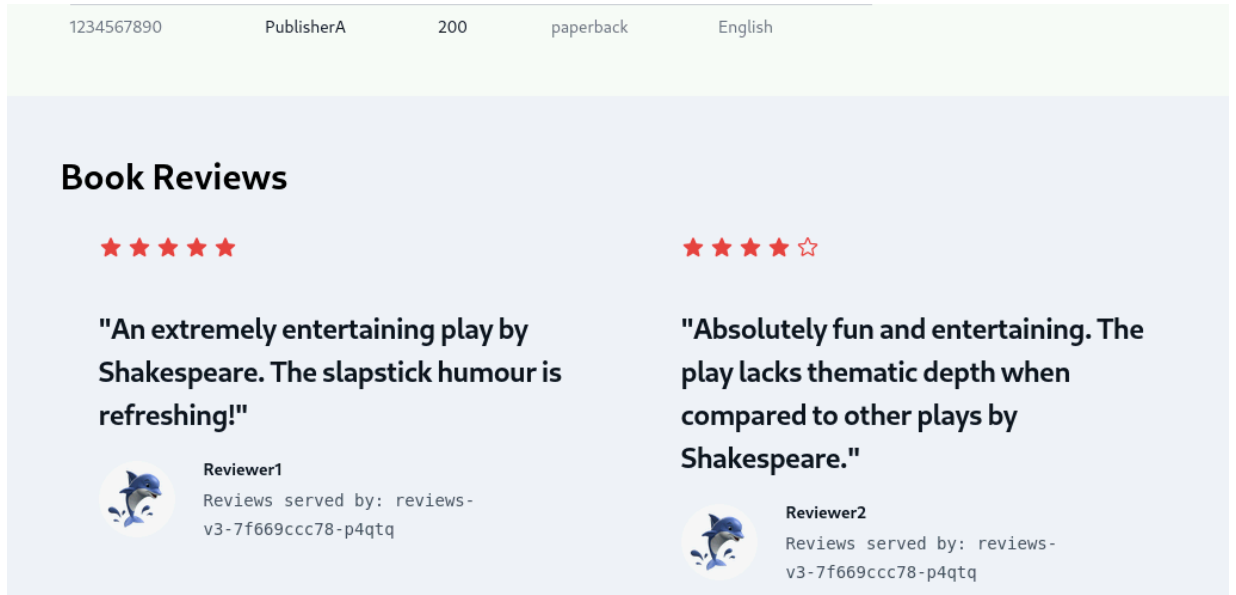


Figure 1.21: Bookinfo application showing only red stars after traffic shifting to v3

Observe that the stars display in red and that the book reviews list the pod serving them as the v3 version of the reviews service. Reload the page a couple of times to ensure it remains unchanged.

- 4.7. Open the terminal window where the `traffic_gen.py` script is running. Press **Ctrl+C** to interrupt the script.

Finish

On the workstation machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish meshintro-bookinfo
```

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